

I. ABSTRACT

ECG signals capture the electrical activity of the heart and can be used to determine its health. The P-QRS-T morphological features of the ECG may be used to identify a variety of cardiac problems. The existence of various disturbances in ECG signals, on the other hand, has a significant impact on correct feature extraction. During acquisition, various noises affect ECG readings, including power line interference, baseline wandering, motion artifacts, and electromyogram noise. In this paper, authors proposed an algorithm called **STationary Wavelet Total Variation (STWaTV)** for denoising ECG signal.

II. OBJECTIVE

- Propose an algorithm named stationary wavelet total variation (**STWaTV**) for the purposes of denoising and artifact removal. This involves introducing the TV method in the SWT domain through the utilization of the bivariate shrinkage rule.
- Test the **STWaTV** algorithm on three existing threshold schemes—firm threshold, hard threshold, and bivariate shrinkage function with SWT—with the aim of determining the optimal threshold rule.
- Evaluate the effectiveness of the proposed **STWaTV** method by conducting a comparative study using standard databases.

III. METHODOLOGY

Denoising signal using STWaTV algorithm

Step1: SWT is applied on noisy ECG signals to get low and high frequency coefficients.

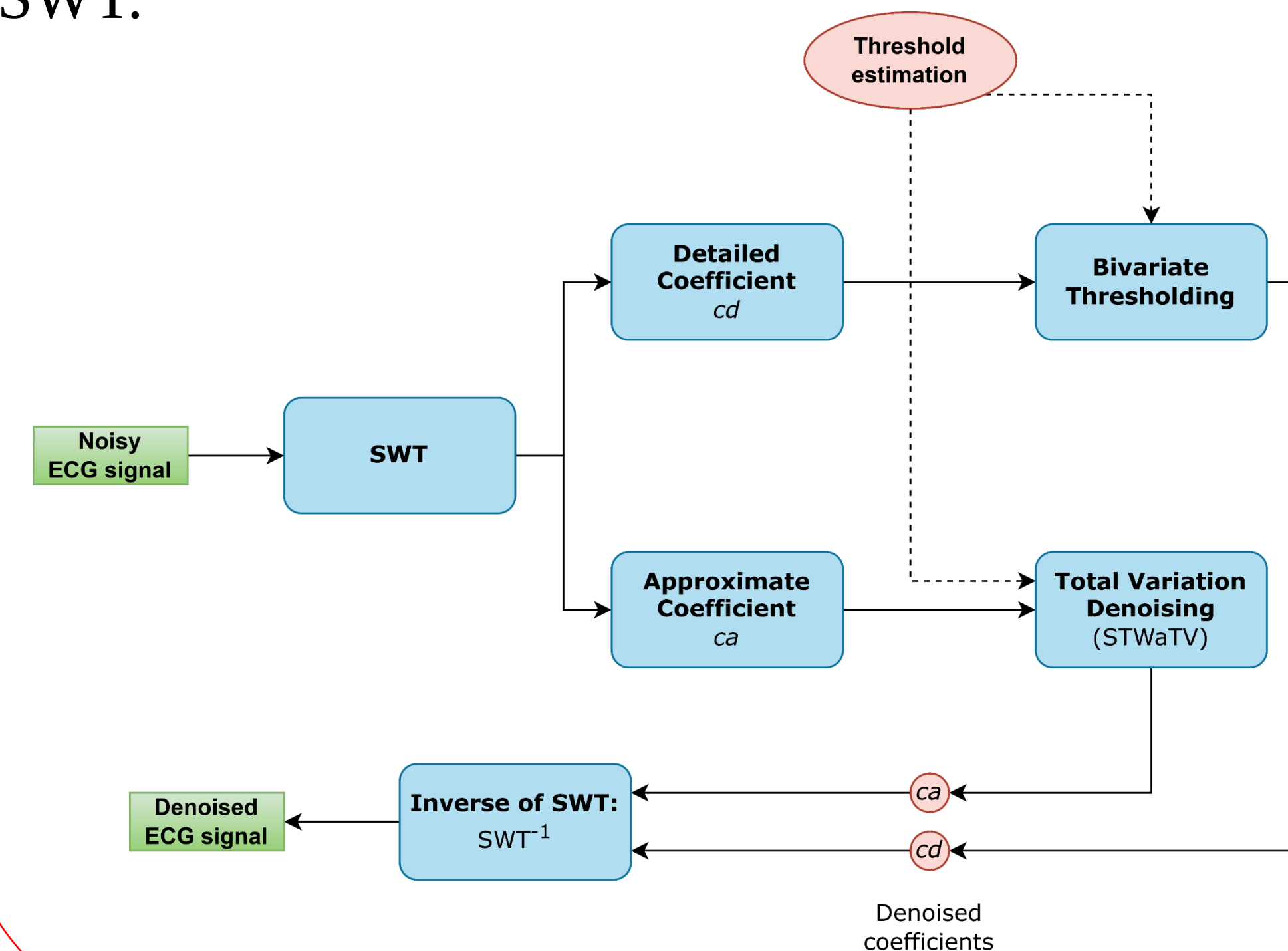
Step2: Threshold value $(\alpha)^+$ estimated using high frequency coefficient through the following equation:

$$\left\{ \begin{array}{l} \hat{\varphi}_1 = \frac{\left(\sqrt{Q_1^2 + Q_2^2} - \sqrt{3} \sigma_n \right)^{\alpha}}{\sqrt{Q_1^2 + Q_2^2}} \times Q_1 \\ \text{denoting : } (\alpha)^+ = \begin{cases} 0 & \alpha < 0 \\ \alpha & \text{else} \end{cases} \end{array} \right.$$

and then high frequency coefficients are thresholded using bivariate shrinkage rule.

Step3: Total variation is applied to approximate coefficients through by minimizing iteratively through Lagrange's method.

Step4: Denoised signal is obtained by inverse of SWT.

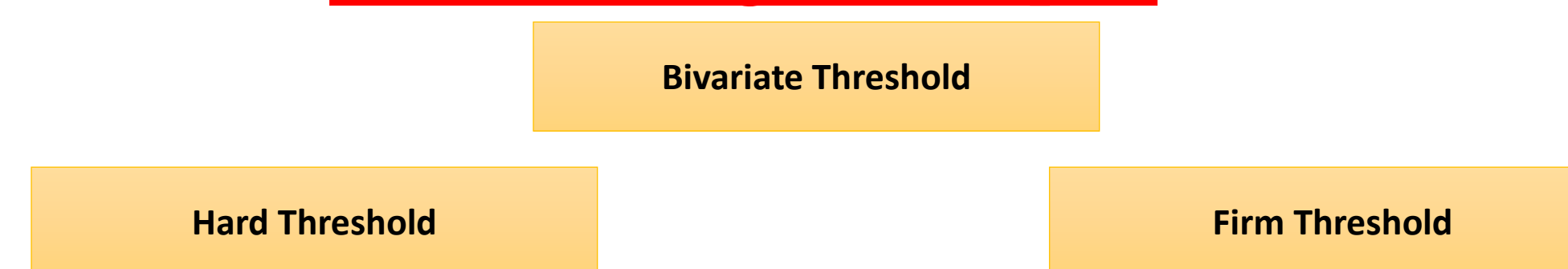


IV. EXPERIMENTAL SETUP

The experimentation is carried out in Matlab2021a.

The proposed STWaTV technique along with various ECG denoising algorithms such as DWT, EMD, FMD, and SWT are examined using three standard ECG databases such as MIT-BIH arrhythmia database, MIT-BIH noise stress test database, and Physionet PTB diagnostic ECG Database.

Thresholding Techniques



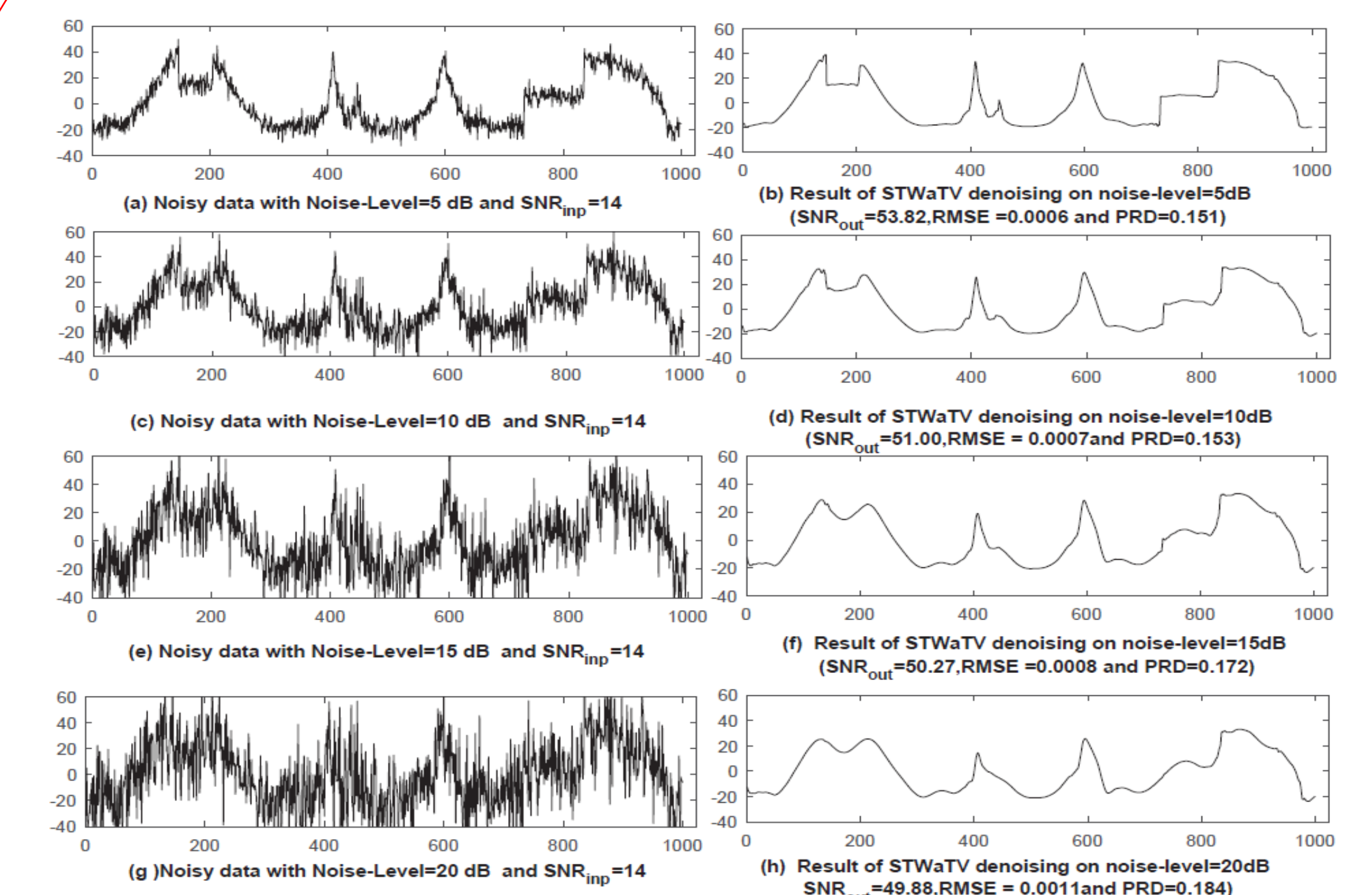
Channel Recordings	Bivariate Theshold	Hard Threshold	Firm Threshold
Average value of Channels	0.0012	0.0021	0.0023

Comparative Study by calculating RMSE of proposed methodology using different Thresholding Schemes.

VI. CONCLUSION

A denoising algorithm for ECG signals relatively free from artifacts is proposed in this work. The TV method is introduced by utilizing the bivariate shrinkage function in conjunction with SWT. The algorithm's efficiency is tested it with three existing thresholds: Bivariate, hard and soft thresholds on three standard ECG databases. Based on the experimental evidence bivariate outperformed as compared to the other two thresholds. Finally, a comparative study has been done along with existing state-of-the-art methods using performance metrics such as SNR, RMSE, and PRD and found the proposed algorithm outperformed with the highest SNR and lowest RMSE and PRD values.

V. RESULT & DISCUSSION



(a)

CHANNEL RECORD	SNR _{IN} (DB)	LPF	HPF	FDM	EMD	DWT	DWT-TV	SWT	STWaTV
Average Value of channels	11.68	25.38	26.72	28.17	33.63	33.83	42.66	44.55	45.47

(b)

CHANNEL RECORD	SNR _{IN} (DB)	LPF	HPF	FDM	EMD	DWT	DWT-TV	SWT	STWaTV
Average Value of channels	11.68	0.1436	0.1676	0.1640	0.1282	0.1280	0.1194	0.1293	0.1075

(c)

CHANNEL RECORD	SNR _{IN} (DB)	LPF	HPF	FDM	EMD	DWT	DWT-TV	SWT	STWaTV
Average Value of channels	11.68	0.443	0.739	0.741	2.359	0.386	0.339	0.341	0.301

Comparative study of proposed methodology for denoising of ECG signals with existing methodologies using (a) SNR, (b) RMSE and (c) PRD.

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